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Oligoclonal CAR-T–Derived CD4+ T-Cell Lymphoproliferative Disorder of the Duodenum Following BCMA-Directed CAR-T Therapy for Multiple Myeloma

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COMPANION SITE
original electropherograms, full references

BACKGROUND

Immune effector cell–associated enterocolitis (IEC-EC) was first described in patients receiving BCMA-directed CAR-T therapy for multiple myeloma, who developed severe non-infectious diarrhea months after infusion (Fortuna 2024). Biopsies showed a duodenum-predominant pattern of injury — villous atrophy, intraepithelial lymphocytosis and loss of plasma cells — that responds to immunosuppression (Osnis 2026; Karimi 2025; Banerjee 2025; Blumenberg 2026). CAR-positive T cells with a cytotoxic phenotype persisted in the mucosa (Ritter 2026; Kethidi 2026), but these lesions were regarded as inflammatory.

Ozdemirli et al. first proposed that the infused product itself could expand clonally, reporting an indolent CD4+ T-cell lymphoma of the small intestine that carried the CAR transgene (2024). It followed the FDA's class-wide warning on T-cell malignancies after CAR-T (Verdun 2024), and further transgene-positive lymphomas were reported, several arising from pre-existing mutated clones or carrying the vector in a tumor suppressor (Perica, Harrison, Braun, Aleman 2025; Shi 2026). Proposed names include *CAR transgenic T-cell lymphoproliferative neoplasm* (Baumgartner 2026) and, for indolent mucosal disease, *indolent intestinal CAR T-cell lymphoproliferative disorder* (Ozdemirli et al. manuscript in revision).

Post-CAR-T gut disease is believed to be a spectrum: non-CAR-mediated injury, polyclonal CAR T-cell enteritis, and clonal CAR T-cell LPD (Somay 2025; Ali 2026; Sarami 2026). Clonal cases carry the most severe enteritis and the highest mortality (Patel 2026), but construct-positive mucosal involvement is not sufficient for a diagnosis of LPD (Susanibar-Adaniya 2026). Diagnostic criteria differentiating CAR T-cell LPD and IEC-EC are currently being developed and include construct-specific immunohistochemistry (camelid V_HH) and serial TCR-γ clonality studies on repeat biopsies.

CLINICAL COURSE

A 74-year-old man with IgG-λ multiple myeloma with *CCND1*::*IGH* received Dara-RVd, a melphalan-conditioned autologous transplant and lenalidomide maintenance, then DKd at relapse. He received ciltacabtagene autoleucel (day 0) after fludarabine–cyclophosphamide lymphodepletion, without cytokine release syndrome or neurotoxicity, and reached an MRD-negative complete remission by clonoSEQ and PET/CT. About three months later he developed progressive diarrhea (more than ten stools daily) and a 20-lb weight loss; infectious studies were negative and the mucosa was endoscopically normal. Duodenal biopsies on days +141 and +190 showed a lamina propria T-cell infiltrate (Panels 01–03). Budesonide, cholestyramine, immunoglobulin and vedolizumab gave partial, transient benefit; ruxolitinib from day +228 restored oral intake. Norovirus with pneumatosis on day +245 led to a negative exploratory laparotomy. A third biopsy on day +303, **after symptomatic improvement**, showed persistent involvement of the duodenum and duodenal bulb. Cyclosporine from day +315, escalated to 450 mg twice daily, resolved the diarrhea; at day +413 he was gaining weight, with normal protein studies and no evidence of myeloma, recurrent diarrhea, or evidence of systemic lymphoma.

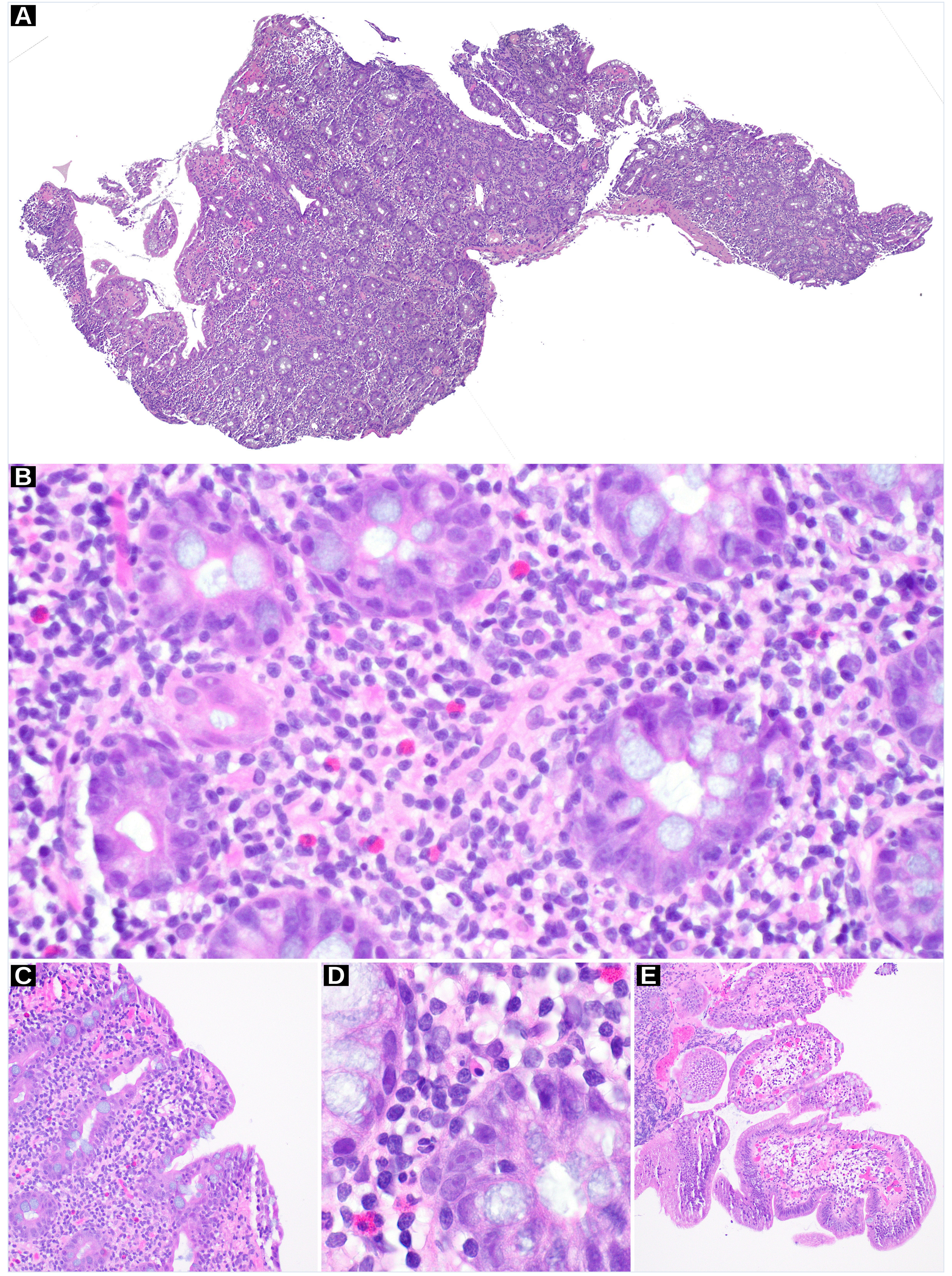
TIMELINE · DAYS FROM INFUSION

DAY 0	Ciltacabtagene autoleucel infused; no cytokine release syndrome or neurotoxicity.
+99	Marrow MRD-negative by clonoSEQ; PET/CT without avid disease.
+141	**First duodenal biopsy** for progressive diarrhea and weight loss; infectious workup negative.
+190	**Second biopsy.** Mucosa endoscopically normal on both occasions.
+193 → +224	Immunoglobulin, then vedolizumab; budesonide and cholestyramine earlier. Partial, transient responses.
+228	Ruxolitinib: symptoms improved, oral intake restored.
+245	Norovirus with pneumatosis; laparotomy for suspected perforation found none.
+303	**Third biopsy,** after symptomatic improvement: duodenum *and* duodenal bulb involved.
+315	Cyclosporine started, later escalated to 450 mg twice daily.
+413	Diarrhea resolved, weight climbing; protein studies normal, myeloma in remission.

ANCILLARY STUDIES

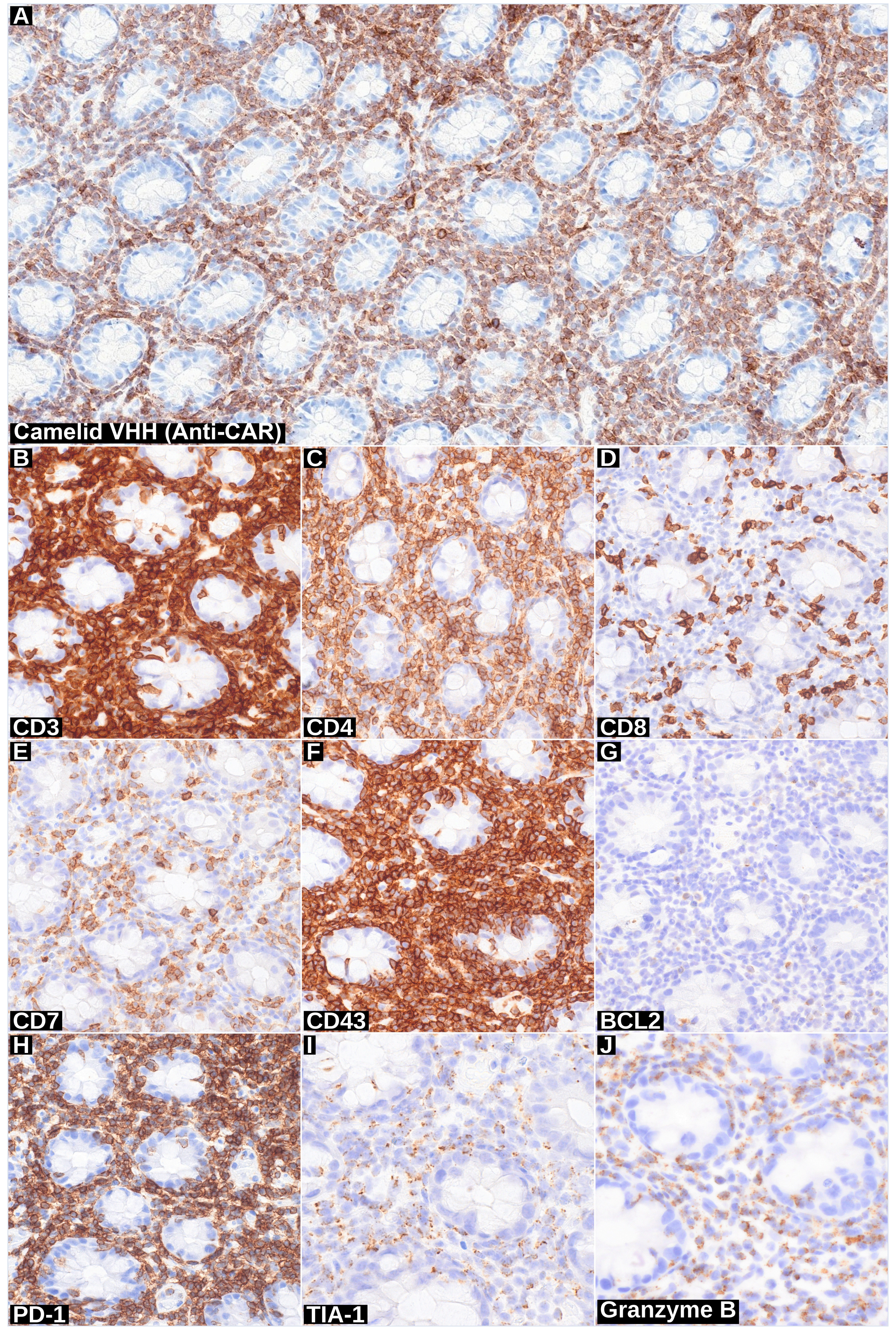
Anti-camelid V_HH	CAR construct in ~70% of infiltrating cells; **robust lamina propria positivity at all three biopsies**
TCR-γ PCR	**Oligoclonal**, the same bands at +141, +190 and +303
Targeted NGS	No relevant variants
TRBC1 flow, blood	No circulating clone
EBER in situ	Negative

01 HISTOPATHOLOGY (DUODENUM; H&E)



A, Low-power overview: a dense lymphocytic infiltrate expands the lamina propria with relative preservation of glandular architecture. **B**, Dense, monotonous lymphoid infiltrate with increased intraepithelial lymphocytes. **C**, Villous blunting with a prominent lamina propria infiltrate. **D**, Apoptotic bodies within glandular epithelium. **E**, Foveolar metaplasia of the surface epithelium.

02 IMMUNOPHENOTYPE



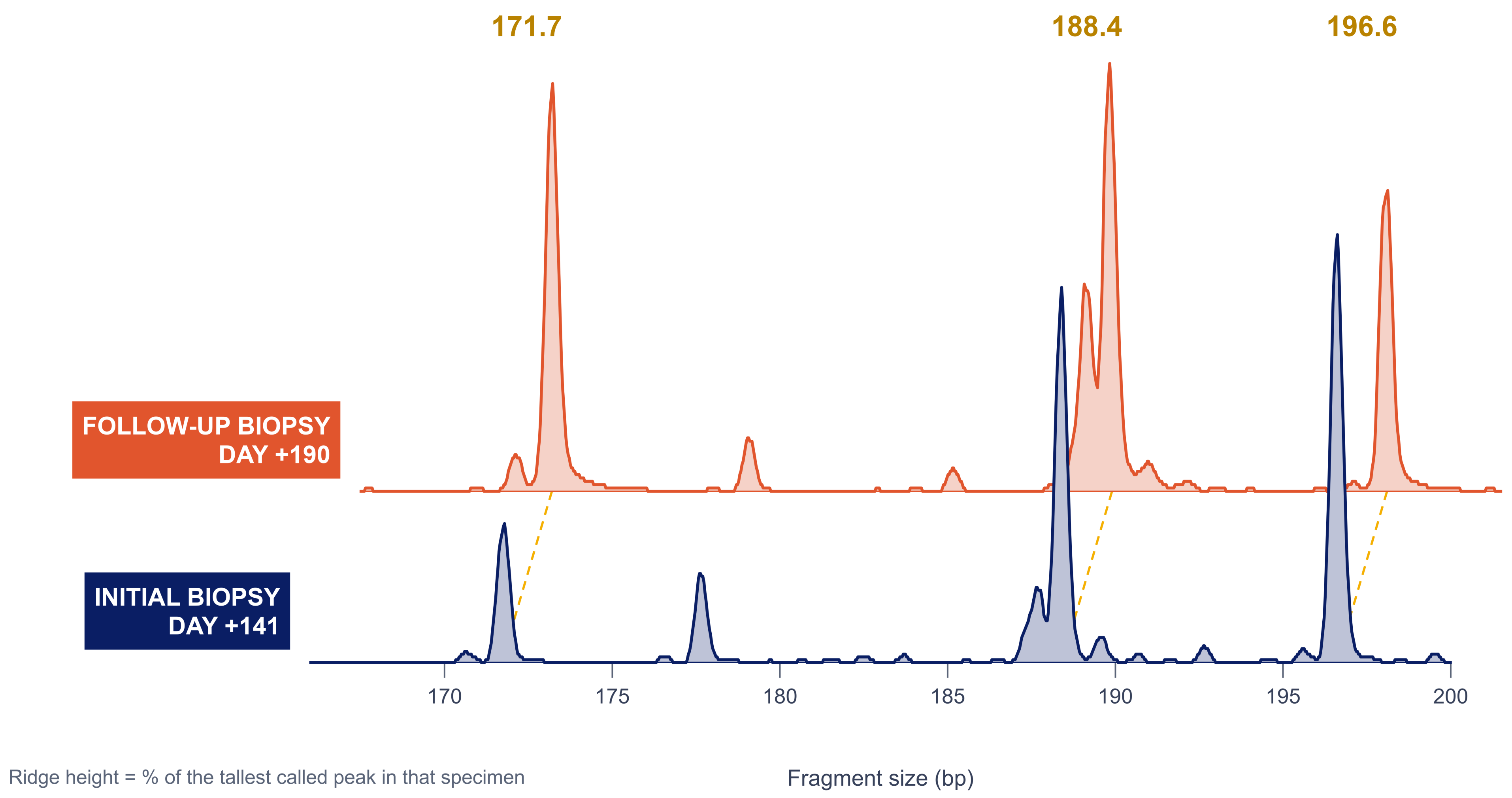
A, Camelid V_HH (anti-CAR) immunohistochemistry confirms construct expression in the majority of infiltrating T cells. **B–D**, CD3 highlights the dense infiltrate; CD4 is positive in most cells, CD8 in a minor subset. **E–G**, CD7 is lost, CD43 diffusely positive, BCL2 negative. **H–J**, PD-1 positive; TIA-1 and granzyme B mark cytotoxic differentiation.

POSITIVE	LOST OR NEGATIVE
CD3 · CD4 · CD5 · CD43 · PD-1 · TIA-1 · granzyme B · camelid V_HH (anti-CAR) · Ki-67 <10%	CD7 (loss) · CD8 (minor subset only) · BCL2 · EBER ISH

Across all three biopsies — days +141, +190 and +303 — the histology, the immunophenotype and the CAR positivity were the same: a dense lamina propria T-cell infiltrate with CD7 loss and robust camelid V_HH staining. **The lesion looked no different in the specimen taken after the symptoms had improved.**

03 TCR-γ CLONALITY ON SERIAL BIOPSIES

TCR-γ multiplex PCR with capillary electrophoresis · re-drawn from the laboratory trace · follow-up biopsy behind, initial biopsy in front on a shared size axis



Ridge height = % of the tallest called peak in that specimen

Gold guides join the three peaks shared by both specimens (171.7, 188.4, 196.6 bp), recurring at the same sizes fifty days apart; 171.7 bp rises to co-dominance at day +190 and a minor 183.6 bp peak appears. **The third biopsy at day +303 reproduced the same bands again. Original instrument electropherograms for all four primer sets are on the companion site.**

05 FEATURES OF IEC-EC AND CLONAL CAR T-CELL LPD

FAVORS IEC-EC

inflammatory pole · 27 patients with tissue

- Apoptosis-dominant, GVHD-like injury — **7 bodies/10 hpf** duodenum, **9.5/10 hpf** ileum
- Crypt dropout and neuroendocrine cell nests in colon
- Foveolar metaplasia
- Infiltrate scattered and mucosa-limited; no expansile, destructive mass
- **CAR-positive cells few or scattered**
- Non-clonal; partial pattern, milder, transient course

PRESENT IN BOTH

the overlap is nearly the whole picture

- Duodenum-predominant injury; colon relatively spared
- Villous atrophy and blunting
- Increased intraepithelial lymphocytes
- Lamina propria T-cell infiltrate
- Plasma cells reduced or absent — **25 of 26 specimens**
- Delayed onset, median **92.5 days**; infection excluded
- Responds to steroids, biologics, ruxolitinib or cyclosporine

FAVORS CAR-T LPD

lymphoproliferative pole · case reports and small series

- Dense, expansile lamina propria infiltrate
- **CAR-positive cells ≥30% of T cells** — ~70% here
- T-cell antigen loss (CD7) despite bland cytology
- Aberrant CD4+ cytotoxic phenotype — TIA-1+, granzyme B+
- **Clonal TCR-γ recurring at identical fragment sizes on serial biopsy**
- Lesion unchanged after the symptoms resolve — this case
- Ki-67 low: a persistent clone, not a proliferative one

04 CONCLUSIONS

- No single biopsy settled this lesion into enterocolitis or lymphoproliferative disorder. It took three specimens over ten months to declare itself.
- **Symptomatic improvement did not clear it.** The same oligoclonal bands, the same aberrant phenotype (CD7 loss; PD-1, TIA-1 and granzyme B on CD4+ T cells) and robust CAR positivity were present at every biopsy, including the one taken after the diarrhea settled.
- It never disseminated, the myeloma stayed in remission, and cyclosporine holds it. The lesion sat between the poles and stayed there.
- **TCR-γ clonality and construct-specific (camelid V_HH) immunohistochemistry belong on every post-CAR-T gastrointestinal biopsy in a myeloma patient** — the standard panel, not a reflex for atypical cases: 2 of 9 IEC-EC workups showed a T-cell LPD pattern (Lim 2026).
- The course fits **indolent intestinal CAR T-cell lymphoproliferative disorder**: localized, construct-positive, clonally stable over serial biopsies, suppressed rather than eradicated by immunosuppression.

References (full citations on the companion site)**:** Ozdemirli 2024 *NEJM* · Fortuna 2024 *Blood Cancer J* · Karimi 2025 *Hum Pathol* · Banerjee 2025 *Blood Cancer J* · Blumenberg 2026 *Blood* · Verdun 2024 *NEJM* · Perica 2025 *NEJM* · Harrison 2025 *NEJM* · Braun 2025 *Nat Med* · Aleman 2025 *NEJM* · Somay 2025 *Blood Cancer J* · Ali 2026 *Nat Rev Clin Oncol* · Baumgartner 2026 *Blood* · Shi 2026 *Blood Adv* · Osnis 2026 *Am J Surg Pathol* · Sarami 2026 *Hum Pathol* · Lim 2026 *Blood Adv* · Ritter 2026 *Leuk Lymphoma* · Patel 2026 *Hum Pathol* · Kethidi 2026 *Nat Med* · Susanibar-Adaniya 2026 *Blood Cancer J* · Hosoya 2026 *Blood*

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